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ELECTRONIC DEVICE, AND FILE SYSTEM OPERATING METHOD OF ELECTRONIC DEVICE

Abstract

An electronic device includes, one or more processors; and memory storing instructions, that, when executed by the one or more processors, cause the electronic device to write files of an application to the memory; determine pinned files; classify hot files and cold files, from among remaining files that are not pinned, according to write patterns of the remaining files; store the pinned files and the cold files in a first area of the memory with a fixed size, and store the hot files in a second area of the memory with a variable size; determine, based on available storage space in the first area, files with changeable allocated block addresses in the first area; and move the files from the first area to the second area, and store the pinned files in a space of the first area that is generated based on moving the files.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a by-pass continuation application of International Application No. PCT/KR2023/018674, filed on Nov. 20, 2023, which is based on and claims priority to Korean Patent Application No. 10-2022-0156437, filed on Nov. 21, 2022, and Korean Patent Application No. 10-2022-0168702, filed on Dec. 6, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND
1. Field
[0002] The present disclosure relates to an electronic device including a memory and a file system operating method of the electronic device.
2. Description of Related Art
[0003] Recently developed electronic devices, such as a smart phone, a tablet PC, a portable multimedia player (PMP), personal digital assistant (PDA), a laptop personal computer, and a wearable device, may perform various functions (e.g., game, social network service (SNS), the Internet, multimedia, and photo and video shooting and execution) as well as mobility.
[0004] The electronic device may include a storage device such as a NAND flash memory or a solid state disk (SSD) to store a large amount of data to perform various functions.
[0005] The electronic device may write at least a part of data of an application to a main memory using a file system, which is a type of program module or software that may be executed by a processor. The electronic device may classify a type of files to be written using the file system at the time of writing the data of the application to the main memory and control the classified files to be separately stored in the memory.
SUMMARY
[0006] According to an aspect of the disclosure, an electronic device includes, one or more processors; and memory storing instructions, that, when executed by the one or more processors, cause the electronic device to write a plurality of files of an application to the memory in response to a file input request of the application; determine one or more pinned files, from among the plurality of files; classify one or more hot files and one or more cold files, from among a remaining plurality of files that are not pinned, according to a plurality of write patterns of the remaining plurality of files; store the one or more pinned files and the one or more cold files in a first area of the memory that has a fixed size, and store the one or more hot files in a second area of the memory that has a variable size, wherein the first area and the second area are separated by a partition; determine, based on available storage space in the first area being less than a predetermined level, one or more first files with allocated block addresses that are changeable, from among a first plurality of files stored in the first area; and move, from the first area to the second area, the one or more first files, and store the one or more pinned files in a space of the first area that is generated based on moving the one or more first files.
[0007] A hot file may be modified or deleted less than a cold file, and a pinned file may have an

allocated block address that is determined to be unchangeable. The instructions, when executed by the one or more processors, may cause the electronic device to identify whether a file is stored in the second area based on receiving, from the application, information indicating that a file block stored in the memory corresponds to the file, and the file is to have an attribute indicating the file is pinned; and move the file to the first area based on identifying that the file is stored in the second area, and designate the attribute to indicate the file is pinned.

[0008] The instructions, when executed by the one or more processors, may cause the electronic device to preferentially store the one or more hot files in the second area; and store at least one of the one or more hot files in the first area based on determining there is insufficient storage space in the second area.

[0009] The instructions, when executed by the one or more processors, may cause the electronic device to preferentially store the one or more cold files in the first area; and store at least one of the one or more cold files in the second area based on determining there is insufficient storage space in the first area.

[0010] The instructions, when executed by the one or more processors, may cause the electronic device to store the one or more pinned files in the first area; and move at least one of the one or more hot files in a first space in the first area to the second area based on determining there is insufficient storage space in the first area, and move at least one of the one or more pinned files into the first space.

[0011] The instructions, when executed by the one or more processors, may cause the electronic device to determine whether no hot files are stored in the first area or whether there would be insufficient storage space available to store the one or more pinned files in the first area after moving the one or more hot files to the second area; and move at least one of the one or more cold files from a second space of the first area to the second area based on determining there is insufficient storage space available to store the one or more pinned files in the first area, and store the one or more pinned files in the second space.

[0012] The one or more hot files may include at least one of a temporary storage file or a cache file, and the one or more cold files may include at least one of a multimedia file or an application execution file.

[0013] The instructions, when executed by the one or more processors, may cause the electronic device to store the plurality of files in the first area, irrespective of attributes of the plurality of files, based on determining the available storage space in the first area exceeds the predetermined level; and store the one or more hot files in the second area based on determining the available storage space in the first area is less than the predetermined level.

[0014] The plurality of write patterns may include at least one of file size, whether a file is modified (dirty page), a time (modification time) when the file was modified, a time interval (modification interval) at which the file was modified, a count (fsync) with which a system called the file, a size (chunk size) of a portion of the file being modified, an extension of the file, a directory name in which the file is stored, or whether a predetermined file system is used.

[0015] The plurality of write patterns may further include an overwrite count indicating a number of times a middle portion of the file is modified, and the instructions, when executed by the one or more processors, may cause the electronic device to determine the number to be frequent based on the overwrite count exceeding a first threshold, and classify the file as a hot file, and determine the number to be infrequent based on the overwrite count being less than a second threshold, and classify the file as a cold file.

[0016] According to an aspect of the disclosure, a non-transitory computer-readable recording medium having instructions recorded thereon, that, when executed by one or more processors, cause the one or more processors to write a plurality of files of an application to memory in response to a file input request of the application; determine one or more pinned files from among the plurality of files; classify one or more hot files and one or more cold files, from among a remaining plurality of files that are not pinned, according to a plurality of write patterns of the plurality of files; store the one or more pinned files and the one or more cold files in a first area of the memory that has a fixed size, and store the one or more hot files in a second area of the memory that has a variable size; determine, based on available storage space in the first area being less than a predetermined level, one or more first files with allocated block addresses that are changeable, from among a first plurality of files stored in the first area; and move, from the first area to the second area, the one or more first files, and store the one or more pinned files in a space of the first area that is generated based on moving the one or more first files.

[0017] A hot file may be modified or deleted less than a cold file, and a pinned file may have an allocated block address that is determined to be unchangeable. The instructions, when executed by the one or more processors, may cause the one or more processors to identify whether a file is stored in the second area based on receiving, from the application, information indicating that a file block stored in the memory corresponds to the file, and the file is to have an attribute indicating the file is pinned; and move the file to the first area based on identifying that the file is stored in the second area, and designate the attribute to indicate the file is pinned.

[0018] The instructions, when executed by the one or more processors, may cause the one or more processors to preferentially store the one or more hot files in the second area; and store at least one of the one or more hot files in the first area based on determining there is insufficient storage space in the second area.

[0019] The instructions, when executed by the one or more processors, may cause the one or more processors to preferentially store the one or more cold files in the first area, and store at least one of the one or more cold files in the second area based on determining there is insufficient storage space in the first area.

[0020] The instructions, when executed by the one or more processors, may cause the one or more processors to store the one or more pinned files in the first area; and move at least one of the one or more hot files stored in a first space in the first area to the second area based on determining there is insufficient storage space in the first area, and move at least one of the one or more pinned files into the first space.

[0021] The instructions, when executed by the one or more processors, may cause the one or more processors to determine whether no hot files are stored in the first area or whether there would be insufficient storage space available to store the one or more pinned files in the first area after moving the one or more hot files to the second area; and move at least one of the one or more cold files inform a second space of the first area to the second area based determining there is insufficient storage space available to store the one or more pinned files in the first area, and store the one or more pinned files in the second space.

[0022] The one or more hot files may include at least one of a temporary storage file or a cache file, and the one or more cold files may include at least one of a multimedia file or an application execution file.

[0023] The instructions, when executed by the one or more processors, may cause the one or more processors to store the plurality of files in the first area, irrespective of attributes of the plurality of files, based on determining the available storage space in the first area exceeds the predetermined level; and store the one or more hot files in the second area based on determining the available storage space in the first area is less than the predetermined level.

[0024] The plurality of write patterns may include at least one of file size, whether a file is modified (dirty page), a time (modification time) when the file was modified, a time interval (modification interval) at which the file was modified, a count (fsync) with which a system called the file, a size (chunk size) of a portion of the file being modified, an extension of the file, a directory name in which the file is stored, or whether a predetermined file system is used.

[0025] The plurality of write patterns may further include an overwrite count indicating a number of times a middle portion of the file is modified, and the instructions, when executed by the one or more processors, may cause the one or more processors to determine the number to be frequent based on the overwrite count exceeding a first threshold, and classify the file as a hot file; and determine the number to be infrequent based on the overwrite count being less than a second threshold, and classify the file as a cold file.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The above and other aspects, features, and advantages of certain embodiments of the present disclosure are more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0027] FIG. **1** is a block diagram of an electronic device in a network environment according to an embodiment.

[0028] FIG. **2** is a diagram illustrating a file classification operation of the electronic device according to an embodiment.

[0029] FIG. **3** is a block diagram illustrating a configuration of the electronic device according to an embodiment.

[0030] FIG. **4**A is a diagram illustrating a situation of shrinking a partition on a file system according to an embodiment.

[0031] FIG. **4**B is a diagram illustrating a file system of the electronic device according to an embodiment.

[0032] FIG. **5** is a diagram illustrating a situation of allocating a file block on the file system of the electronic device according to an embodiment.

[0033] FIG. **6** is a diagram illustrating a situation of removing the file block on the file system of the electronic device according to an embodiment.

[0034] FIG. **7** is a flowchart of a file system operating method of the electronic device according to an embodiment.

DETAILED DESCRIPTION

[0035] The embodiments described in the disclosure, and the configurations shown in the drawings, are only examples of embodiments, and various modifications may be made without departing from the scope and spirit of the disclosure.

[0036] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0037] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0038] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0039] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0040] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0041] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0042] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0043] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0044] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0045] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0046] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0047] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0048] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0049] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0050] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0051] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0052] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0053] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of Ims or less) for implementing URLLC.

[0054] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0055] According to various embodiments, the antenna module **197** may form a mm Wave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mm Wave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0056] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0057] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0058] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0059] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0060] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0061] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.
[0062] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
[0063] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.
[0064] FIG. **2** illustrates a file classification operation of the electronic device according to an embodiment.
[0065] The electronic device (e.g., the electronic device **101** of FIG. **1**) may include a processor (e.g., the processor **120** of FIG. **1**) and a memory **220**. The processor **120** may classify and store files in the memory **220** in response to a file write request of at least one application. The processor **120** may classify files as either hot files or cold files according to pre-defined rules **201** on a file system **225** in the memory **220**. The hot files may mean files that are relatively frequently modified and/or deleted compared to the cold files. The cold files may mean files that are relatively infrequently modified and/or deleted compared to the hot files. The hot file may include, for example, DB, xml files, and the cold file may include, for example, media files such as photo and video. The pre-defined rules **201** may include at least one of a rule set according to a method of predicting the possibility that files will be changed based on, for example, file extensions (.xml,.db,.jpg,.mp4, for example), a rule set according to a method of predicting the possibility that files will be changed based on directory characteristics specified for each OS, or a rule set according to a method of predicting the possibility that files will be changed based on a block write size, for example.

[0066] The file system **225** may mean a program module (e.g., the program module **140** of FIG. **1**) that may be executed by the processor **120**. The electronic device may separately store a plurality of files in the memory **220** using the file system **225**.
[0067] The processor **120** may identify whether the file classification information and the file data on the application match the file data stored in the memory **220** using a writeback thread module **203** of the file system **225**, and when the file data in the memory **220** does not match the file data on the application, control to perform the writeback.
[0068] The writeback may mean an operation of updating only a cache of the memory **220** when writing the file data. The processor **120** may control to update only the cache when writing the file data, and write the file data to a main memory device or an auxiliary memory device including the memory **320**.
[0069] The electronic device **101** may use a write pattern of at least a part of a time period between the time when the files are written to a page cache in the memory **220** and the time when the files are stored in the memory **220** to separately store files (hot files) that are frequently modified and deleted and files (cold files) that are infrequently changed after storage.
[0070] The electronic device **101** may classify the type of files using the file system **225** and separately store the files in the memory **220**.
[0071] The electronic device **101** may use the write pattern from the time when the files are written to the page cache to the time when the files are stored in the storage to separately store the files (hot files) that are frequently modified and deleted and the files (cold files) that are infrequently changed after the file storage.
[0072] The technology for separately storing the hot files and the cold files are being handled by various methods in the fields of the memory **220** and the file system **225**. The reason is that, when separately storing the hot files and the cold files, the NAND storage or the file system may refer to the information to minimize a garbage collection operation, or when storing the cold files with a relatively larger size compared to the hot files, data blocks are continuously stored to minimize fragmentation, thereby optimizing the file storage performance and the NAND storage lifespan.
[0073] The electronic device **101** may support some files (e.g., pinned files) with file attributes of which the block movement is restricted. An application may manage a location where file data blocks are stored within the application. The application may directly transmit read and write requests to a block address of a storage device without going through the file system for the read and write requests of the data block. When the data block addresses of the pinned files are arbitrarily changed on the file system **225**, a problem of accessing an incorrect data address on the application may occur. To prevent this problem, the electronic device **101** may store the pinned files in a space where the pinned files are not reclaimed and control the data block addresses of the pinned files not to be arbitrarily changed on the file system **225**.
[0074] FIG. **3** is a block diagram illustrating a configuration of the electronic device according to an embodiment.
[0075] Referring to FIG. **3**, an electronic device **300** may include a processor **310** and a memory **320**, and some of the illustrated components may be substituted. The electronic device **300** may further include at least some of the components and/or functions of the electronic device **101** of FIG. **1**. At least some of the components of the illustrated (or not illustrated) electronic device may be operatively, functionally, and/or electrically connected to each other.
[0076] According to an embodiment, the processor **310** is a component capable of performing calculations or data processing related to control and/or communication of each component of the electronic device **300**, and may be composed of one or more processors. The processor **310** may include at least some of the components and/or functions of the processor **120** of FIG. **1**.
[0077] According to an embodiment, the calculation and data processing functions that may be implemented by the processor **310** on the electronic device **300** are not limited. Characteristics related to the control of the file system **325** in the memory **320** will be described in detail. The

operations of the processor **310** may be performed by loading the instructions stored in the memory **320** (e.g., the memory **130** of FIG. **1**). The file system **325** structure may mean a structure for storing or organizing files or data so that the electronic device **300** may discover and access the files or data. The file system **325** according to an embodiment may configure the file system structure to store and manage files and directories in the memory **320**.

[0078] At least one memory **320** may include the volatile memory **132** of FIG. **1**, and may also include the nonvolatile memory **134** of FIG. **1**. The memory **320** may include a volatile memory such as a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous dynamic RAM (SDRAM). The memory **320** may include at least one of one time programmable ROM (OTPROM), PROM, EPROM, EEPROM, mask ROM, flash memory, hard drive, or solid state drive (SSD). The memory **320** is the non-volatile memory and may include a large capacity storage device. For example, the memory **320** may include at least one of one time programmable ROM (OTPROM), PROM, EPROM, EEPROM, mask ROM, flash ROM, hard drive, or solid state drive (SSD). The memory **320** may store various file data, and the stored file data may be updated according to the operation of the processor **310**.

[0079] According to an embodiment, the file system **325** may mean a system that can be used by intercepting a system call at a kernel level without a separate daemon. The daemon may mean a background process that is executed when the system is first operated. The file system **325** may wait for the user's request while the daemon is stored in the memory **320** and then recognize the user's request when the user's request occurs.

[0080] According to an embodiment, the file system **325** may provide files in response to a request from an application layer. The file system **325** may be located on a kernel layer. The operation of providing the files may mean an operation of opening and reading the requested files or reading the already open files to transmit the read open files to the application layer. The operation of opening the files may mean an operation of finding names of the files in the storage device and preparing to read and/or write the files in the application layer. The operation of reading the files may mean an operation of loading the open file data into the memory **320**. The loading may mean an operation of fetching a program itself and resources for the operation from the auxiliary memory device (e.g., hard disk) to the main memory device (e.g., memory).

[0081] According to an embodiment, the file system **325** may read the requested file and provide the files to the application layer in response to the request of the application layer when the requested file exists on an upper file system in the memory **320**, and read the files on the nonvolatile memory and provide the files to the application layer when the requested files do not exist on the upper file system.

[0082] According to an embodiment, the processor **310** may execute various types of software (e.g., the program **140**). The memory **320** may include the file system **325**. The file system **325** is a program module stored in the memory **320** and may be operated by the processor **310**.

[0083] According to an embodiment, the processor **310** may store data in the form of the files in the memory **320** through the file system **325**. The file system **325** may mean a data structure or system that the processor **310** manages to store data in the memory **320**. The electronic device **300** may use the file system **325** to write data to the memory **320** or efficiently read the data stored in the memory **320**. In an embodiment, the file system **325** is described assuming that the file system **325** is implemented as a flash friendly file system (F2FS), but the form of the file system **325** is not limited to the F2FS, and may include other forms of file systems. The F2FS may mean a file system optimized for a NAND flash memory based on log-based storage.

[0084] According to an embodiment, the memory **320** may include a first area **331** in which a first file or the cold files are stored. The memory **320** may include a second area **332** in which a second file or the hot files are stored. The first file or the cold files may mean files that are relatively infrequently modified and/or deleted compared to the hot files. The second file or the hot files may mean files that are relatively frequently modified and/or deleted compared to the cold files.

[0085] According to an embodiment, the file system **325** may include a writeback thread module (e.g., the writeback thread module **203** of FIG. **2**). The writeback thread module **203** may detect a request from an application to write files to the file system **325** and monitor key characteristics (hereinafter referred to as a 'write pattern' or 'first pattern') of the write operation. The writeback thread module **203** may temporarily store the write pattern of the file being monitored in a file object (not illustrated) in the memory **320**. The write pattern may include at least one of the file size, whether the files are modified (dirty page), a time (modification time) when the files are modified, a time interval (modification interval) at which the files are modified, a count (fsync) with which the system calls the files, a size (chunk size) of a portion being modified on the files, extensions of the files, directory names in which the files are stored, and whether a predetermined file system is used.
[0086] According to an embodiment, features of the write patterns of the files may include at least one of an overwrite count, an append count, a write chunk, or a system call count (fsync).
[0087] The overwrite count may mean the number of times some of the files are modified in the middle. For example, the processor **310** may classify the files as the overwrite count when storing the files in the case where some cells in an Excel file are modified. When the overwrite count of the files is relatively large, the processor **310** may determine that the files are frequently modified and classify the files as being close to the hot files.
[0088] For example, when the overwrite count is greater than a first threshold, the processor **310** may determine that the files are frequently modified and may classify the files as being hot files. When the overwrite count is less than a second threshold, the processor **310** may determine that the files are infrequently modified and may classify the files as being cold files.
[0089] The append count may mean the number of times other data is newly added to the existing files. For example, the processor **310** may classify files as the append count when storing the files in the case where new data is added beyond the level of the files being partially modified in a jpg file. When the append count of the files is relatively large, the processor **310** may determine that new contents are often added to the files and may classify the files as being close to the cold files.
[0090] The write chunk may mean a file size unit stored in an operation of writing the files to the memory **320**. For example, when the write chunk is relatively small, the processor **310** may determine that a small number of writes are performed in a one-time write operation and classify the files as being close to the hot files. Conversely, when the write chunk is relatively large, the processor **310** may determine that a large number of writes are performed in a one-time write operation and classify the files as being close to the cold files.
[0091] The system call count (fsync) may mean the number of times the files recorded in the cache of the memory **320** are written to the non-volatile memory. For example, when the fsync is lower than a predetermined value, the processor **310** may determine that the files recorded in the memory **320** are files that are written relatively less frequently. In other words, when the fsync is lower than the predetermined value, the processor **310** may classify the files as the cold files that are infrequently modified. Conversely, when the fsync is higher than the predetermined value, the processor **310** may classify the files as the hot files that are relatively frequently modified.
[0092] According to an embodiment, the processor **310** may classify whether files are the hot files or the cold files based on the features of at least one write pattern described above.
[0093] FIG. **4**A is a diagram illustrating a situation of shrinking a partition on the file system.
[0094] Referring to FIG. **4**A, the electronic device (e.g., the electronic device **300** of FIG. **3**) may support some files (e.g., pinned files) with the file attribute of which the block movement is restricted. The electronic device **300** may divide an area on the memory (e.g., the memory **320** of FIG. **3**) and store files in different areas according to their attributes. A first area (e.g., the first area **331** of FIG. **3**) **401** and a second area (e.g., the second area **332** of FIG. **3**) **402** may be divided by a partition on the memory **320**. According to an embodiment, the electronic device **300** may generate a new partition or support a shrink operation of some partitions to expand spaces of other

partitions. The electronic device **300** may reclaim some areas (e.g., the second area **402**) to shrink the partition. The reclaimed space (e.g., the second area **402**) may be an available space where the files are not stored. The files stored in the reclaimed space (e.g., the second area **402**) may be deleted after the reclamation operation of the electronic device **300**. The electronic device **300** may move the files stored in the reclaimed space (e.g., the second area **402**) to the non-reclaimed space (e.g., the first area **401**) to prevent files from being deleted or lost. The movement of the files may incur costs, and when the number of files moved increases, the files are deleted during the movement of the files or are left in the space that is reclaimed without being moved, so the number of files being deleted may increase.

[0095] The electronic device **300** and the file system operating method of the electronic device **300** according to an embodiment may reduce the number of files stored in the reclaimed space (e.g., the second area **402**) when shrinking the partition, and reduce costs due to the movement of the files. The electronic device **300** according to an embodiment may mainly arrange the files (e.g., the hot files) expected to be deleted soon according to the file attribute in the reclaimed space (e.g., the second area **402**), and control the data loss to be minimized even if there are the files that are left in the reclaimed space and deleted. The file system operating method of the electronic device for minimizing the costs and data loss due to the movement of the files will be described.

[0096] FIG. **4**B is a diagram illustrating the file system of the electronic device according to an embodiment.

[0097] The first area **401** may mean a non-reclaimable extent when shrinking the partition. The processor (e.g., the processor **310** of FIG. **3**) may preferentially store the pinned files and the cold files in the first area **401**. The pinned files may mean the files with the file attributes of which the block movement is restricted. The cold files may mean the files that are relatively infrequently modified and/or deleted compared to the hot files. The cold file may include, for example, media files such as photo and video. This is only an example, and the type of cold files is not limited thereto, and the cold files may include any file that is infrequently modified and/or deleted.

[0098] The memory **320** may include the non-reclaimable first area **401** and the reclaimable second area **402**. The second area **402** may further include a plurality (e.g., N) of areas **402_1**, **402_2**, . . . , **402**_N-**1**, and **402**_N. The areas **402_1**, **402_2**, . . . , **402**_N-**1**, and **402**_N within the second area **402** may have priorities in terms of the reclamation. For example, when shrinking the partition, the 2-Nth area **402**_N may be reclaimed first, and the 2-1th area **402_1** may be reclaimed last. The number or priority of areas within a reclaimable extent is not fixed and may vary depending on the settings.

[0099] FIG. **5** is a diagram illustrating a situation of allocating a file block on the file system of the electronic device according to an embodiment.

[0100] The processor (e.g., the processor **310** of FIG. **3**) may receive a write request for pinned files **505** from an application and store the pinned files **505** in the memory (e.g., the memory **320** of FIG. **3**). Since the pinned files **505** have the file attributes of which the block movement is restricted, the processor **310** may allocate the pinned files to the non-reclaimable first area **401**. The processor **310** may allocate the pinned files **505** to the non-reclaimable first area **401**, thereby reducing costs due to the movement of the files and preventing the loss of the pinned files **505** that may occur during the movement of the files. The application may designate files as the pinned files based on determining to obtain the attributes of the pinned files.

[0101] According to an embodiment, the processor **310** may identify whether there is a sufficient space to store the pinned files **505** in the first area **401**. Based on the determination that there is the insufficient space to store the pinned files **505** in the first area **401**, the processor **310** may move some files stored in the first area **401** to the second area **402** to secure the storage space. The second area **402** means a reclaimable area in the operation of shrinking the partition. The processor **310** may move the hot files **507**, which are relatively frequently modified and/or deleted compared to the cold files, to the second area **402**. The processor **310** may store the pinned files **505** in the space generated while moving the hot files on the first area **401**.

[0102] According to an embodiment, the processor **310** may identify whether the files are stored in the second area **402** based on receiving, from the application, a signal or information indicating that the file block stored in the memory **320** has the attributes corresponding to the pinned files **505**. The processor **310** may move the files to the first area **401** and designate the attributes corresponding to the pinned files **505** based on identifying that the files are stored in the second area **402**. When the files stored in the reclaimable second area **402** are changed to have the attributes of the pinned files, the processor **310** may move the files to the first area **401** in the operation of reclaiming the partition later. Since the movement of the files incur costs and poses a risk of the file loss, the processor **310** may move the files to the first area **401** in advance and designate the attributes corresponding to the pinned files **505**.

[0103] According to an embodiment, the processor **310** may preferentially store the hot files in the second area **402**, and store the hot files in the first area **401** based on the insufficient storage space in the second area **402**.

[0104] According to an embodiment, unlike the first area **401**, the second area **402** may experience the area shrink due to the partition shrink. In the process of shrinking the second area **402**, the movement or deletion of the files may occur. It may be advantageous for the processor **310** not to store the files in the second area **402** in terms of reducing costs or the risk of the file loss. For this reason, the processor **310** may store files of all attributes in the first area **401** until the available space in the first area **401** becomes less than the predetermined level (e.g., 20%), and allocate some files (e.g., the hot files **507**) with attributes to the second area **402** on the basis that the available space in the first area **401** becomes less than the predetermined level. The predetermined level (e.g., 20%) is only an example and is not fixed, and may vary according to the settings.

[0105] According to an embodiment, the processor **310** may preferentially store the cold files in the first area **401**, and store the cold files in the second area **402** based on the insufficient storage space in the first area **401**.

[0106] The processor **310** may store the pinned files **505** in the first area **401**, move at least some of the hot files **507** stored in the first area **401** to the second area **402** based on the insufficient storage space in the first area **401**, and store the pinned files **505** in the space in the first area **401** in which the hot files **507** are stored.

[0107] The processor **310** may identify whether there is no hot files stored in the first area **401** or whether there is the insufficient available space to store the pinned files **505** in the first area **401** even after moving the hot files **507** stored in the first area **401**, move at least some of the cold files stored in the first area **401** to the second area **402** based on the insufficient available space to store the pinned files **505** in the first area **401**, and store the pinned files **505** in the space on the first area **401** in which the cold files are stored.

[0108] FIG. **6** is a diagram illustrating a situation of removing the file block on the file system of the electronic device according to an embodiment.

[0109] FIG. **610** illustrates a situation in which files (e.g., hot files **630** and cold files **632**) are stored without dividing between the non-reclaimable first area **401** and the reclaimable second area **402**. In the FIG. **610**, the cold files **632** may be stored in an area with a high reclamation priority on the second area **402**. The cold files **632** on the area being shrunk may be deleted in the partition shrink situation. The processor (e.g., the processor **310** of FIG. **3**) may move the cold files **632** to the non-reclaimable first area **401** to prevent the deletion of the cold files **632**. Costs may be incurred during the movement of the files. Some or all of the cold files **632** being moved may be lost in the situation where the available space in the first area **401** is insufficient.

[0110] The electronic device (e.g., the electronic device **300** of FIG. **3**) and the file system operating method of the electronic device **300** according to an embodiment may reduce costs and the risk of file loss during the movement of the files. This will be described in FIG. **620**. In the FIG. **620**, the processor **330** may classify the files based on the file attributes, and control cold files **632** to be stored in the first area **401** and hot files **630** to be stored in the second area **402**. The electronic device **300** according to an embodiment may control the files not to be stored in the reclaimable second area **402**, or control the hot files **630**, which have a high probability of being deleted or invalidated within a short period of time, to be stored. When shrinking the partition, only valid files (or file blocks) may be a movement target. The valid files may include, for example, the cold files **632** or the pinned files (e.g., the pinned files **505** of FIG. **5**). The electronic device **300** according to an embodiment may control not to store the cold files **632** or the pinned files **505** in the second area **402**. Compared to the FIG. **610**, the electronic device **300** according to an embodiment may store only invalid files (e.g., the hot files **630**), which are not file move targets, in the second area **402** to reduce can reduce the number of files (or the number of file blocks) moved when shrinking the partition.

[0111] FIG. **7** is a flowchart of a file system operating method of the electronic device according to an embodiment.

[0112] The operations described with reference to FIG. **7** may be implemented based on instructions that may be stored in a computer recording medium or a memory (e.g., the memory **320** of FIG. **3**). An illustrated method **700** may be executed by the electronic device (e.g., the electronic device **300** of FIG. **3**) described with reference to FIGS. **1** to **6** above, and reference may be made to the descriptions of FIGS. **1** to **6** for additional implementation details. The order of each operation of FIG. **7** may be changed, and some operations may be performed simultaneously.

[0113] In operation **710**, the processor (e.g., the processor **310** of FIG. **3**) may determine whether the files are the pinned file while writing the application file on the memory (e.g., the memory **320** of FIG. **3**). The pinned files may mean the files with the file attributes of which the block movement is restricted. The application may designate the files as the pinned files based on determining to obtain the attributes of the pinned files. The processor **310** may identify the attributes of the files to determine whether the files requested to be written to the application have the attributes of the pinned files.

[0114] In operation **720**, the processor **310** may classify the files as either the hot files or the cold files according to the write patterns of the files. The processor **310** may classify the files as either the hot files or the cold files according to the write patterns of the files on the basis that the files requested to be written to the application do not have the attributes of the pinned files. The hot files may mean the files that are relatively frequently modified and/or deleted compared to the cold files. The cold files may mean the files that are relatively infrequently modified and/or deleted compared to the hot files. The write pattern may include at least one of the file size, whether the files are modified (dirty page), the time (modification time) when the files are modified, the time interval (modification interval) at which the files are modified, the count (fsync) with which the system calls the files, the size (chunk size) of a portion being modified on the files, the extensions of the files, the directory names in which the files are stored, and whether the file system is used. According to an embodiment, the features of the write patterns of the files may include at least one of the overwrite count, the append count, the write chunk, or the system call count (fsync).

[0115] The overwrite count may mean the number of times some of the files are modified in the middle. For example, the processor **310** may classify the files as the overwrite count when storing the files in the case where some cells in the Excel file are modified. When the overwrite count of the files is relatively large, the processor **310** may determine that the files are frequently modified and classify the files as being close to the hot files.

[0116] For example, when the overwrite count is greater than a first threshold, the processor **310** may determine that the files are frequently modified and may classify the files as being hot files. When the overwrite count is less than a second threshold, the processor **310** may determine that the files are infrequently modified and may classify the files as being cold files.

[0117] The append count may mean the number of times other data is newly added to the existing files. For example, the processor **310** may classify the files as the append count when storing the files in the case where new data is added beyond the level of the files being partially modified in the jpg file. When the append count of the files is relatively large, the processor **310** may determine that new contents are often added to the files and may classify the files as being close to the cold files.

[0118] The write chunk may mean the file size unit stored in the operation of writing the files to the memory **320**. For example, when the write chunk is relatively small, the processor **310** may determine that a small number of writes are performed in a one-time write operation and classify the files as being close to the hot files. Conversely, when the write chunk is relatively large, the processor **310** may determine that a large number of writes are performed in a one-time write operation and classify the files as being close to the cold files.

[0119] The system call count (fsync) may mean the number of times the files recorded in the cache of the memory **320** are written to the non-volatile memory. For example, when the fsync is relatively low, the processor **310** may determine that the files recorded in the cache of the memory **320** are the files that are written relatively few times. When the fsync is relatively low, the processor **310** may determine that the files are infrequently modified and classify the files as being close to the cold files. Conversely, when the fsync is relatively high, the processor **310** may determine that the files are relatively frequently modified and classify the files as being close to the hot files. When the fsync is less than a first level, the processor **310** may determine that the files are infrequently modified and classify the files as being close to the cold files. For example, when the fsync is less than 3, the processor **310** may determine that the files are infrequently modified and classify the files as being close to the cold files. The value or the first level of the fsync may not be limited thereto, and may be determined by developer's settings or may be determined by training multiple files through machine learning.

[0120] According to an embodiment, the processor **310** may classify whether the files are the hot files or the cold files based on the features of at least one write pattern described above.

[0121] In operation **730**, the processor **310** may store the pinned files and the cold files in the first area (e.g., the first area **401** of FIG. **4**A) of which the size is fixed, and store the hot files in the second area (e.g., the second area **402** of FIG. **4**A) of which the size may vary. The first area **401** may mean a non-reclaimable extent when shrinking the partition. The processor (e.g., the processor **310** of FIG. **3**) may preferentially store the pinned files and the cold files in the first area **401**. The cold file may include, for example, media files such as photo and video. This is only an example, and the type of cold files is not limited thereto, and the cold files may include any file that is infrequently modified and/or deleted.

[0122] According to an embodiment, the processor **310** may preferentially store the hot files in the second area **402**, and store the hot files in the first area **401** based on the insufficient storage space in the second area **402**.

[0123] In operation **740**, the processor **310** may determine the files of which the allocated block address is changeable among the files stored in the first area **401** based on identifying that there is no available space in the first area **401**.

[0124] According to an embodiment, unlike the first area **401**, the second area **402** may experience the area shrink due to the partition shrink. In the process of shrinking the second area **402**, the movement or deletion of the files may occur. It may be advantageous for the processor **310** not to store the files in the second area **402** in terms of reducing costs or the risk of the file loss. For this reason, the processor **310** may store files of all attributes in the first area **401** until the available space in the first area **401** becomes less than the predetermined level (e.g., 20%), and allocate some files (e.g., the hot files) with attributes to the second area **402** on the basis that the available space in the first area **401** becomes less than the predetermined level. The predetermined level (e.g., 20%) is only an example and is not fixed, and may vary according to the settings.

[0125] According to an embodiment, based on the determination that there is the insufficient space to store the pinned files in the first area **401**, the processor **310** may move some files stored in the first area **401** to the second area **402** to secure the storage space. The second area **402** means the reclaimable area in the operation of shrinking the partition. The processor **310** may move the hot files, which are relatively frequently modified and/or deleted compared to the cold files, to the second area **402**.

[0126] In operation **750**, the processor **310** may move the files of which the block addresses are changeable from the first area to the second area, and store the pinned files in the space in the first area, generated by the movement of the files of which the block addresses are changeable. The processor **310** may store the pinned files **505** in the space generated while moving the hot files on the first area **401**.

[0127] The electronic device may include the memory including the first area and the second area separated by the partition and the processor. The processor may write the files of the application to the memory in response to the file input request of the application, determine whether the files are the pinned files, classify the files as either the hot files or the cold files according to the write patterns of the files on the basis that the files are not the pinned files, store the pinned files and the cold files in the first area of which the size is fixed and store the hot files in the second area of which the size varies, determine, on the basis that the available space in the first area is less than the predetermined level, the files of which the allocated block addresses are changeable from among the files stored in the first area, and move, from the first area to the second area, the files of which the block addresses are changeable and store the pinned files in the space in the first area, generated by the movement of the files of which the block addresses are changeable.

[0128] The hot files are the files that are frequently modified and/or deleted compared to the cold files, the cold files are the files that are infrequently modified and/or deleted compared to the hot files, the pinned files means the files of which the allocated block addresses are determined unchangeable, and the processor may identify whether the files are stored in the second area based on receiving the signal or information from the application indicating that the file block stored in the memory has the attributes corresponding to the pinned files, move the files to the first area based on identifying that the files are stored in the second area, and designate the attributes corresponding to the pinned files. The processor may preferentially store the hot files in the second area, and store the hot files in the first area based on the insufficient storage space in the second area. The processor may preferentially store the cold files in the first area, and store the cold files in the second area based on the insufficient storage space in the second area.

[0129] The processor may store the pinned files in the first area, and move at least some of the hot files stored in the first area to the second area based on the insufficient storage space in the first area, and store the pinned files in the space in the first area in which the hot files are stored.

[0130] The processor may identify whether there is no hot files stored in the first area or whether there is the insufficient available space to store the pinned files in the first area even after moving the hot files stored in the first area, move, based on the insufficient available space to store the pinned files in the first area, at least some of the cold files stored in the first area to the second area, and store the pinned files in the space in the first area in which the cold files are stored.

[0131] The hot files may include temporary storage files or cache files, and the cold files may include multimedia files or application executable files.

[0132] The processor may store all the files of the attributes in the first area on the basis that the size of the available space in the first area exceeds the predetermined level, and store the hot files in the second area on the basis that the size of the available space in the first area is less than the predetermined level.

[0133] The write pattern may include at least one of the file size, whether the files are modified (dirty page), the time (modification time) when the files are modified, the time interval (modification interval) at which the files are modified, the count (fsync) with which the system calls the files, the size (chunk size) of a portion being modified on the files, the extensions of the files, the directory names in which the files are stored, and whether the predetermined file system is

used.
[0134] The write pattern may include the overwrite count which means the number of times some of the files are modified in the middle, and when the overwrite count of the files is relatively large, the processor may determine that the files are frequently modified and classify the files as being close to the hot files, and when the overwrite count of the files is relatively small, the processor may determine that the number of times the files are modified is small and classify the files as being close to the cold files.
[0135] For example, when the overwrite count is greater than a first threshold, the files may be determined to be frequently modified and may be classified as hot files. When the overwrite count is less than a second threshold, the files may be determined to be infrequently modified and may be classified as cold files.
[0136] The various embodiments merely provide examples to facilitate an understanding of the disclosure and are not intended to limit the scope of the disclosure. The scope of the disclosure should be interpreted as including all changes or modifications derived based on the technical idea of an embodiment, in addition to the embodiments disclosed herein.

Claims

1. An electronic device, comprising: one or more processors; and memory storing instructions, wherein the instructions, when executed by the one or more processors, cause the electronic device to: write a plurality of files of an application to the memory in response to a file input request of the application; determine one or more pinned files, from among the plurality of files; classify one or more hot files and one or more cold files, from among a remaining plurality of files that are not pinned, according to a plurality of write patterns of the remaining plurality of files; store the one or more pinned files and the one or more cold files in a first area of the memory that has a fixed size, and store the one or more hot files in a second area of the memory that has a variable size, wherein the first area and the second area are separated by a partition; determine, based on available storage space in the first area being less than a predetermined level, one or more first files with allocated block addresses that are changeable, from among a first plurality of files stored in the first area; and move, from the first area to the second area, the one or more first files, and store the one or more pinned files in a space of the first area that is generated based on moving the one or more first files.
2. The electronic device of claim 1, wherein a hot file is modified or deleted less than a cold file, and a pinned file has an allocated block address that is determined to be unchangeable, and wherein the instructions, when executed by the one or more processors, cause the electronic device to: identify whether a file is stored in the second area based on receiving, from the application, information indicating that a file block stored in the memory corresponds to the file, and the file is to have an attribute indicating the file is pinned; and move the file to the first area based on identifying that the file is stored in the second area, and designate the attribute to indicate the file is pinned.
3. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, cause the electronic device to: preferentially store the one or more hot files in the second area; and store at least one of the one or more hot files in the first area based on determining there is insufficient storage space in the second area.
4. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, cause the electronic device to: preferentially store the one or more cold files in the first area; and store at least one of the one or more cold files in the second area based on determining there is insufficient storage space in the first area.
5. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, cause the electronic device to: store the one or more pinned files in the first area; and move at least one of the one or more hot files in a first space in the first area to the second area

based on determining there is insufficient storage space in the first area, and move at least one of the one or more pinned files into the first space.

6. The electronic device of claim 5, wherein the instructions, when executed by the one or more processors, cause the electronic device to: determine whether no hot files are stored in the first area or whether there would be insufficient storage space available to store the one or more pinned files in the first area after moving the one or more hot files to the second area; and move at least one of the one or more cold files from a second space of the first area to the second area based on determining there is insufficient storage space available to store the one or more pinned files in the first area, and store the one or more pinned files in the second space.

7. The electronic device of claim 1, wherein the one or more hot files comprise at least one of a temporary storage file or a cache file, and wherein the one or more cold files comprise at least one of a multimedia file or an application execution file.

8. The electronic device of claim 1, wherein the instructions, when executed by the one or more processors, cause the electronic device to: store the plurality of files in the first area, irrespective of attributes of the plurality of files, based on determining the available storage space in the first area exceeds the predetermined level; and store the one or more hot files in the second area based on determining the available storage space in the first area is less than the predetermined level.

9. The electronic device of claim 1, wherein the plurality of write patterns comprise at least one of: file size, whether a file is modified (dirty page), a time (modification time) when the file was modified, a time interval (modification interval) at which the file was modified, a count (fsync) with which a system called the file, a size (chunk size) of a portion of the file being modified, an extension of the file, a directory name in which the file is stored, or whether a predetermined file system is used.

10. The electronic device of claim 9, wherein the plurality of write patterns further comprise an overwrite count indicating a number of times a middle portion of the file is modified, and wherein the instructions, when executed by the one or more processors, cause the electronic device to: determine the number to be frequent based on the overwrite count exceeding a first threshold, and classify the file as a hot file, and determine the number to be infrequent based on the overwrite count being less than a second threshold, and classify the file as a cold file.

11. A non-transitory computer-readable recording medium having instructions recorded thereon that, when executed by one or more processors, cause the one or more processors to: write a plurality of files of an application to memory in response to a file input request of the application; determine one or more pinned files from among the plurality of files; classify one or more hot files and one or more cold files, from among a remaining plurality of files that are not pinned, according to a plurality of write patterns of the plurality of files; store the one or more pinned files and the one or more cold files in a first area of the memory that has a fixed size, and store the one or more hot files in a second area of the memory that has a variable size; determine, based on available storage space in the first area being less than a predetermined level, one or more first files with allocated block addresses that are changeable, from among a first plurality of files stored in the first area; and move, from the first area to the second area, the one or more first files, and store the one or more pinned files in a space of the first area that is generated based on moving the one or more first files.

12. The non-transitory computer-readable recording medium of claim 11, wherein a hot file is modified or deleted less than a cold file, and a pinned file has an allocated block address that is determined to be unchangeable, and wherein the instructions, when executed by the one or more processors, cause the one or more processors to: identify whether a file is stored in the second area based on receiving, from the application, information indicating that a file block stored in the memory corresponds to the file, and the file is to have an attribute indicating the file is pinned; and move the file to the first area based on identifying that the file is stored in the second area, and designate the attribute to indicate the file is pinned.

13. The non-transitory computer-readable recording medium of claim 11, wherein the instructions, when executed by the one or more processors, cause the one or more processors to: preferentially store the one or more hot files in the second area; and store at least one of the one or more hot files in the first area based on determining there is insufficient storage space in the second area.
14. The non-transitory computer-readable recording medium of claim 11, wherein the instructions, when executed by the one or more processors, cause the one or more processors to: preferentially store the one or more cold files in the first area, and store at least one of the one or more cold files in the second area based on determining there is insufficient storage space in the first area.
15. The non-transitory computer-readable recording medium of claim 11, wherein the instructions, when executed by the one or more processors, cause the one or more processors to: store the one or more pinned files in the first area; and move at least one of the one or more hot files stored in a first space in the first area to the second area based on determining there is insufficient storage space in the first area, and move at least one of the one or more pinned files into the first space.
16. The non-transitory computer-readable recording medium of claim 15, wherein the instructions, when executed by the one or more processors, cause the one or more processors to: determine whether no hot files are stored in the first area or whether there would be insufficient storage space available to store the one or more pinned files in the first area after moving the one or more hot files to the second area; and move at least one of the one or more cold files inform a second space of the first area to the second area based determining there is insufficient storage space available to store the one or more pinned files in the first area, and store the one or more pinned files in the second space.
17. The non-transitory computer-readable recording medium of claim 11, wherein the one or more hot files comprise at least one of a temporary storage file or a cache file, and wherein the one or more cold files comprise at least one of a multimedia file or an application execution file.
18. The non-transitory computer-readable recording medium of claim 11, wherein the instructions, when executed by the one or more processors, cause the one or more processors to: store the plurality of files in the first area, irrespective of attributes of the plurality of files, based on determining the available storage space in the first area exceeds the predetermined level; and store the one or more hot files in the second area based on determining the available storage space in the first area is less than the predetermined level.
19. The non-transitory computer-readable recording medium of claim 11, wherein the plurality of write patterns comprise at least one of: file size, whether a file is modified (dirty page), a time (modification time) when the file was modified, a time interval (modification interval) at which the file was modified, a count (fsync) with which a system called the file, a size (chunk size) of a portion of the file being modified, an extension of the file, a directory name in which the file is stored, or whether a predetermined file system is used.
20. The non-transitory computer-readable recording medium of claim 19, wherein the plurality of write patterns further comprise an overwrite count indicating a number of times a middle portion of the file is modified, and wherein the instructions, when executed by the one or more processors, cause the one or more processors to: determine the number to be frequent based on the overwrite count exceeding a first threshold, and classify the file as a hot file; and determine the number to be infrequent based on the overwrite count being less than a second threshold, and classify the file as a cold file.